

Secret of the Polyglot

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

The Network Operations Center (NOC) of your local institution has detected a suspicious file. The challenge: the file claims to be one type (PDF), but the system is reporting conflicting information about its actual nature. Your mission is to identify the true file type and extract the hidden flag by examining the file from multiple perspectives.

This is a classic example of a **polyglot file** - a file that can be interpreted as multiple different file types depending on how it's opened or analyzed.

Hints

- The file extension can be misleading
 - File headers (magic bytes) reveal the true file type
 - Different tools may interpret the file differently
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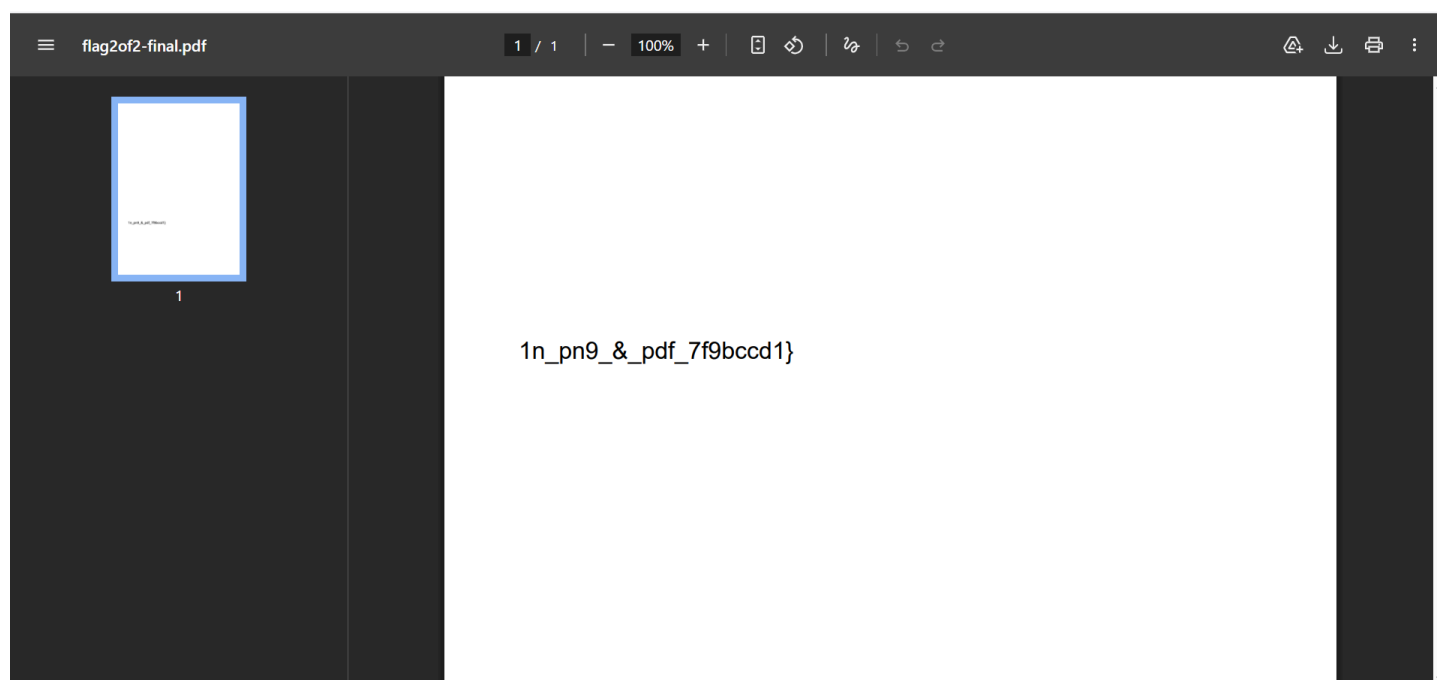
Tools Used

- **file:** Identify the actual file type using magic bytes
 - **strings:** Extract readable text strings from binary data
 - **grep:** Search for patterns (like "picoCTF" flag format)
 - **exiftool:** Examine metadata embedded in files
 - **convert:** Transform files between different formats (ImageMagick)
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Complete Solution

Step 1: Download and Prepare

Download the suspicious file named `flag2of2-final.pdf` from the challenge page.



Step 2: File Type Detection

The key to solving polyglot challenges is identifying the **true file type**, not relying on the file extension. Use the `file` command, which examines the file's magic bytes (header signature):

```
file flag2of2-final.pdf
```

Expected output:

```
flag2of2-final.pdf: PNG image data, 50 x 50, 8-bit/color RGBA, non-interlaced
```

Critical observation: Despite the `.pdf` extension, the `file` command reveals this is actually a **PNG image**, not a PDF. This is the core of the polyglot challenge-the file masquerades as one type but is actually another.

Step 3: Search for Embedded Text

The flag might be stored directly within the file as readable text. Extract all printable strings and filter for the picoCTF flag format:

```
strings flag2of2-final.pdf | grep -i "picoCTF"
```

⚠️ *If no results appear, the flag is encoded within the image data itself.*

Step 4: Examine Metadata

Use `exiftool` to inspect embedded metadata that might contain clues or the flag:

```
exiftool flag2of2-final.pdf
```

Complete output:

```
ExifTool Version Number      : 12.76
File Name                    : flag2of2-final.pdf
Directory                   : .
File Size                    : 3.4 kB
File Modification Date/Time   : 2026:05:08 15:37:32+01:00
File Access Date/Time        : 2026:05:08 15:37:32+01:00
File Inode Change Date/Time   : 2026:05:08 15:37:52+01:00
File Permissions              : -rwxrwxrwx
File Type                    : PNG
File Type Extension           : png
MIME Type                    : image/png
Image Width                  : 50
Image Height                  : 50
Bit Depth                    : 8
Color Type                   : RGB with Alpha
Compression                   : Deflate/Inflate
Filter                       : Adaptive
Interlace                    : Noninterlaced
Profile Name                  : ICC profile
Profile CMM Type              : Little CMS
Profile Version               : 4.3.0
Profile Class                 : Display Device Profile
Color Space Data              : RGB
Profile Connection Space      : XYZ
Profile Date Time             : 2023:11:02 17:42:31
Profile File Signature        : acsp
Primary Platform              : Apple Computer Inc.
CMM Flags                     : Not Embedded, Independent
Device Attributes             : Reflective, Glossy, Positive, Color
Rendering Intent              : Perceptual
Connection Space Illuminant   : 0.9642 1 0.82491
Profile Creator               : Little CMS
Profile ID                    : 0
Profile Description           : GIMP built-in sRGB
Profile Copyright             : Public Domain
Media White Point             : 0.9642 1 0.82491
Chromatic Adaptation         : 1.04788 0.02292 -0.05022 0.02959 0.99048 -0.01707
                              -0.00925 0.01508 0.75168
Red Matrix Column             : 0.43604 0.22249 0.01392
```

Blue Matrix Column	: 0.14305 0.06061 0.71393
Green Matrix Column	: 0.38512 0.7169 0.09706
Red Tone Reproduction Curve	: (Binary data 32 bytes, use -b option to extract)
Green Tone Reproduction Curve	: (Binary data 32 bytes, use -b option to extract)
Blue Tone Reproduction Curve	: (Binary data 32 bytes, use -b option to extract)
Chromaticity Channels	: 3
Chromaticity Colorant	: Unknown
Chromaticity Channel 1	: 0.64 0.33002
Chromaticity Channel 2	: 0.3 0.60001
Chromaticity Channel 3	: 0.15001 0.06
Device Mfg Desc	: GIMP
Device Model Desc	: sRGB
Pixels Per Unit X	: 11811
Pixels Per Unit Y	: 11811
Pixel Units	: meters
Modify Date	: 2023:11:02 17:57:06
Comment	: Created with GIMP
Warning	: [minor] Trailer data after PNG IEND chunk
Image Size	: 50x50
Megapixels	: 0.003

Key findings:

- **File Type:** PNG (not PDF as the extension suggests)
- **Dimensions:** 50x50 pixels (small image-likely a QR code or simple graphic)
- **Color Type:** RGB with Alpha (supports transparency)
- **Comment:** Created with GIMP

Step 5: Convert and Extract the Image

Now that we know the true file type is PNG, we have two approaches:

Option A - Rename the file:

Simply rename the file from `.pdf` to `.png` :

```
mv flag2of2-final.pdf flag2of2-final.png
```

Option B - Use ImageMagick convert:

Transform the file format explicitly:

```
convert flag2of2-final.pdf flag2of2-final.png
```

Either method reveals the PNG image:

Step 6: Extract the Flag from the Image

Open the resulting PNG image with an image viewer or examine its pixel data. The image will contain the flag visually encoded or embedded within the image metadata/data.

Final Result

picoCTF{<REDACTED>}

Key Concepts

- **Polyglot Files:** Files that can be validly interpreted as multiple different file types, exploiting how applications parse file formats.
- **File Headers (Magic Bytes):** The first few bytes of a file that identify its true type, independent of the filename extension.
- **File Extension Unreliability:** Extensions can be misleading; always verify with tools like `file` command.
- **Metadata Analysis:** Tools like exiftool reveal embedded information that can aid in file identification and forensic analysis.
- **Multiple Interpretation Methods:** Different tools may interpret the same file differently based on their parsing logic.
- **Practical Polyglot Example:** This challenge demonstrates how a PNG image can be renamed with a PDF extension to deceive casual inspection.